

Express Train

// Express

Goal

Set up and use the `express.js` web application framework in NodeJS

Background

As our websites become more and more complex, writing our back-end using the `http` module becomes more and more cumbersome and complicated.

In this topic, we'll see an alternative module: **Express.js**.

Express is a **web application framework** for NodeJS. It's designed to help us easily build the back-end for our web applications by providing easy to use solutions for the most common tasks.

In express, we start out by creating our app:

```
const express = require('express');  
const app = express();
```

Using this `app` object we can begin to define our endpoints and request handlers:

```
app.get('/', (req, res) => {  
  res.send("Hello from express!")  
})
```

This defines a GET request handler for the `/` endpoint. This is called a **"route"**. To add another endpoint, all we have to do is define more routes:

```
app.get('/page1', (req, res) => {  
  res.sendFile("page1.html")  
})
```

Notice the `res` object is different from the one used in the `http` module. Using the `res` object we can customize the response sent to the user. Among other things we can:

1. Send some text using `res.send`. This will automatically set the Status to `200 OK`!
2. Send the contents of a file using `res.sendFile` (simpler than manually reading the file).
3. Send an HTTP response code using `res.sendStatus(404)`.
4. Redirect the client to another route: `res.redirect('/profile')`

Finally, like before, we need to start up the server and listen for incoming clients:

```
const PORT = 80;  
  
app.listen(PORT, () => {  
  console.log(`App is listening on ${PORT}`)  
})
```

Description

Use the attached `index.js` and run the code.

Create or add the following routes:

1. The `/` endpoint will serve an HTML page
2. The `/joke` endpoint will send a message with a funny joke

3. The `/print` endpoint will cause the server to log "I was told to print this" to the console and respond with 200 OK
4. The `/redirect` endpoint will cause the server to redirect the client to the `/print` endpoint

To Submit

Submit your code in a javascript file named `index.js`.

